# Engineering Change Notice(ECN)

## ECN-D-2607-01(Single Robot 12inch pusher P/N update)

**FA Research ME 2 team**

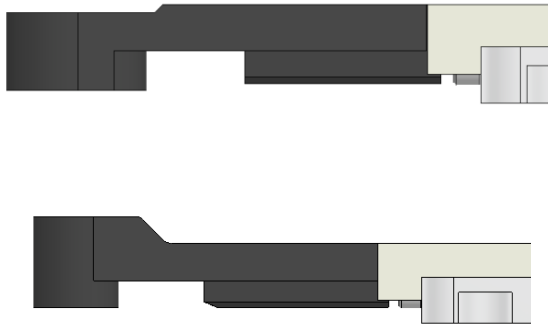
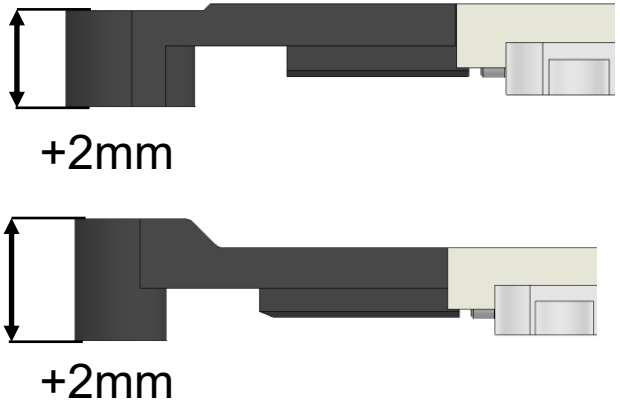
| Written | Review | Confirm |
|---|---|---|
| 김문선 | - | 최동수 |
| 07/03 | - | 07/03 |

NINEBELL

# 1. Engineering Change Request

N/A

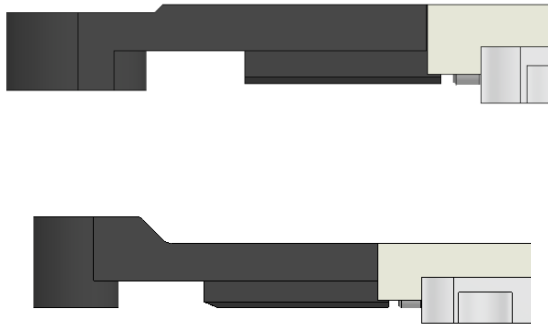
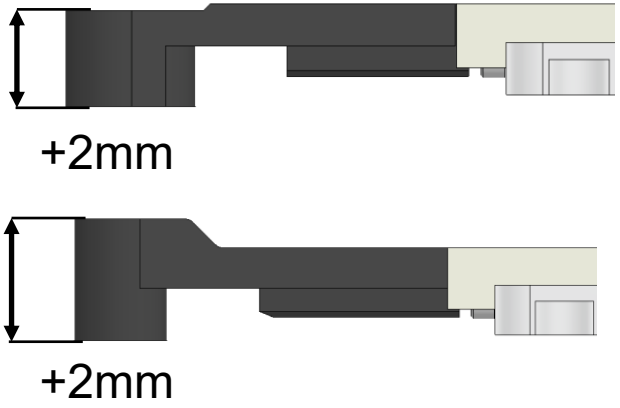
# 2. Engineering Change Notice

| | | | |
|---|---|---|---|
| 적용 장비 타입<br>Applicable project type | SINGLE ROBOT 12" PUSHER | 적용 범위<br>Scope | ■ 신규 발주분 (New order)<br>□ 기발주분 (Previous order)<br>□ 기출하장비 (Existing shipped equipment) |
| 변경 사유<br>Reason for change | □ 고객 요청 (Customer request) ■ 품질개선 (Quality improve)<br>□ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity)<br>■ 부품 단종 (Part discontinuation) □ 기타 (ETC) | ECR NO.<br>ECN NO. | N/A<br>ECN-D-2607-01 |
| 변경 상세 사유<br>Detailed reason for change | 푸셔 웨이퍼 클램프 시 웨이퍼 상단 하단 마진 Spec이 나인벨 내부적으로 결정한 Spec 범위에 만족하지 않음.<br>RCS, 메일로 공유된 12인치 푸셔 품번에 대해 모두 리비전하여 업데이트.<br>추후 PO, CS, SPARE 진행 시 -0 품번을 사용을 방지 하기 위함. | | |
| 변경 사항<br>Changes | 변경 전(Before) | 변경 후(After) | |
| |  |  | |
| 상세내용<br>Detail | DR002-PS12-M004-0 & DR002-PS12-M005-0 : PUSHER TIP<br>DR002-PS12-ESAC-0 : STD A/C PUSHER<br>DR002-PS12-ESBD-0 : STD B/D PUSHER<br>DR002-PS12-ESAC-5WP0 : 5mm Warpage A/C PUSHER<br>DR002-PS12-ESBD-5WP0 : 5mm Warpage B/D PUSHER | DR002-PS12-M004-1 & DR002-PS12-M005-1 : PUSHER TIP<br>DR002-PS12-ESAC-1 : STD A/C PUSHER<br>DR002-PS12-ESBD-1 : STD B/D PUSHER<br>DR002-PS12-ESAC-5WP1 : 5mm Warpage A/C PUSHER<br>DR002-PS12-ESBD-5WP1 : 5mm Warpage B/D PUSHER | |



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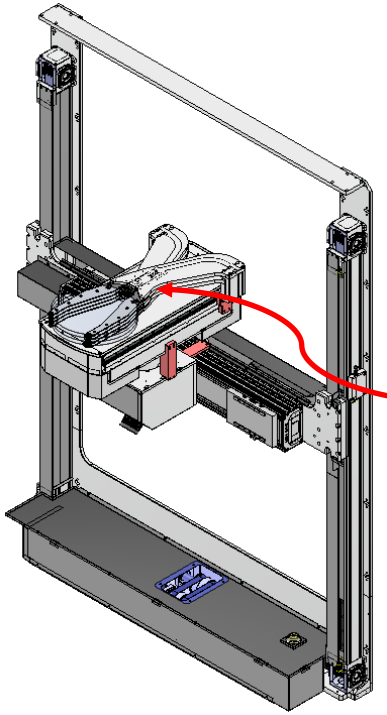
# 2. Engineering Change Notice ENG

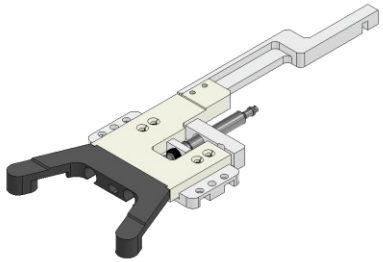
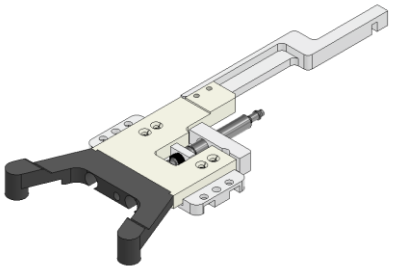
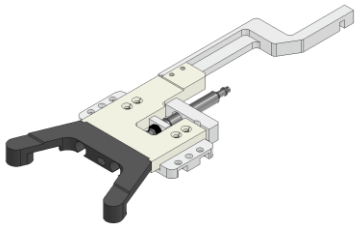
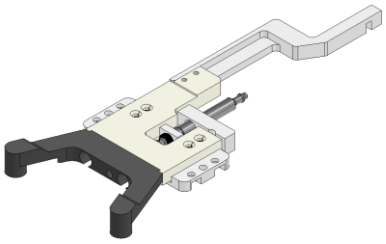
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|---|---|---|---|
| 적용 장비 타입<br>Applicable project type | SINGLE ROBOT 12″ PUSHER | 적용 범위<br>Scope | ■ 신규 발주분 (New order)<br>□ 기발주분 (Previous order)<br>□ 기출하장비 (Existing shipped equipment) |
| 변경 사유<br>Reason for change | □ 고객 요청 (Customer request) ■ 품질개선 (Quality improve)<br>□ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity)<br>■ 부품 단종 (Part discontinuation) □ 기타 (ETC) | ECR NO.<br>ECN NO. | N/A<br>ECN-D-2607-01 |
| 변경 상세 사유<br>Detailed reason for change | **- The wafer top and bottom margin specifications during pusher wafer clamping do not meet the internal specifications established by NINEBELL.**<br>**- Revise and update all 12-inch pusher part numbers shared via email in the RCS.**<br>**- This is to prevent the use of '-0' part numbers in future PO, CS, and SPARE processing.** | | |
| 변경 사항<br>Changes | 변경 전(Before) | 변경 후(After) | |
| |  |  | |
| 상세내용<br>Detail | DR002-PS12-M004-0 & DR002-PS12-M005-0 : PUSHER TIP<br>DR002-PS12-ESAC-0 : STD A/C PUSHER<br>DR002-PS12-ESBD-0 : STD B/D PUSHER<br>DR002-PS12-ESAC-5WP0 : 5mm Warpage A/C PUSHER<br>DR002-PS12-ESBD-5WP0 : 5mm Warpage B/D PUSHER | DR002-PS12-M004-1 & DR002-PS12-M005-1 : PUSHER TIP<br>DR002-PS12-ESAC-1 : STD A/C PUSHER<br>DR002-PS12-ESBD-1 : STD B/D PUSHER<br>DR002-PS12-ESAC-5WP1 : 5mm Warpage A/C PUSHER<br>DR002-PS12-ESBD-5WP1 : 5mm Warpage B/D PUSHER | |



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# 2. Engineering Change Notice

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| 전체 이미지<br>Full Image |  <br>※변경 위치 혹은 관련 위치 표시<br>※Show changes or related positions | | | | | |
| 유효성 검토 결과<br>Test Result | 검토 여부<br>Test or not | Y / N | 유효성 검토 부서<br>Department | N/A | 고객 승인 여부<br>Customer approva | Y / N |
| 승인 날짜<br>Approval date | 1. Ninebell : 2026.07.03<br>2. Customer : / / | | 나인벨 승인<br>Ninebell approval | QC 2026.07.03 PASS | 고객사 승인<br>Customer approval | ACM confirms<br>Lei Wu Single RobotT<br>2026/07/07 |
| 적용 일자<br>Effective date | After approval | | 최초 적용 장비<br>Initially Applied Equipment | After approval | | |

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|---|---|
|  |  |
| DR002-PS12-ESAC-1 | DR002-PS12-ESAC-5WP1 |
|  |  |
| DR002-PS12-ESBD-1 | DR002-PS12-ESBD-5WP1 |

# 3. Attachments

N/A